From grasp pose to joint motion

MoveIt 2 connects a geometric grasp target to an executable robot trajectory.

01 ALIGN THE TARGET

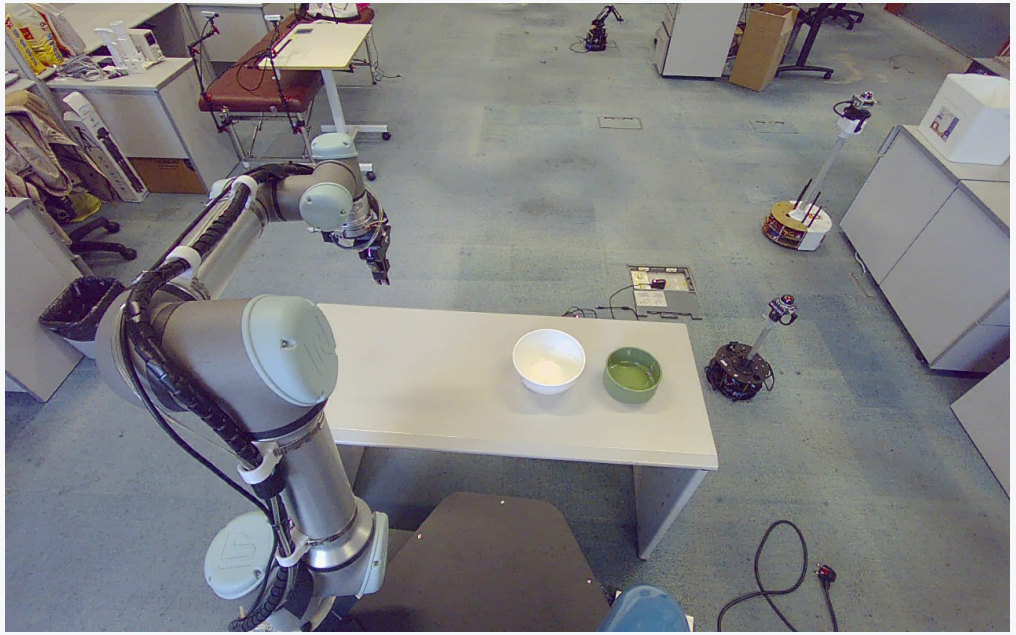
Convert grasp_center to tool0.
Use measured joints as the starting state.

02 SOLVE INVERSE KINEMATICS

Find joint angles that realize the tool pose.
One pose may have several arm configurations.

03 PREFER NEARBY JOINT SOLUTIONS

Rank valid IK solutions by joint displacement.
Up to 8 search attempts; plan up to 3 solutions.



Original laboratory scene · 27 Sep 2026

$FK(q) = T_{target}$

IK ranking: $J = \Sigma_i (q_i - q_{i,current})^2$

IK: joint angles. Planning: a route.

q: six joint angles T: tool position + orientation | Sources: M1, M3; image: I1

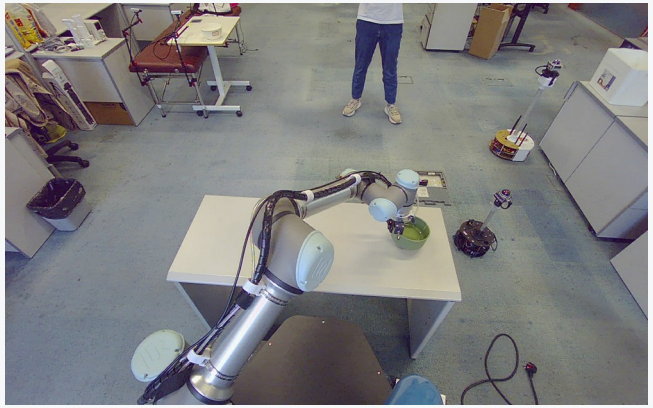
Two planning methods, one six-segment task

Joint-space search for transfer; Cartesian interpolation for vertical contact motions.

SEGMENT		METHOD
1	Approach: 10 cm above grasp	**RRTConnect**
2	Descend to rim	**Cartesian line**
3	Lift: +10 cm	**Cartesian line**
4	Transfer to fixed ready pose	**RRTConnect**
5	Lower: −20 cm	**Cartesian line**
6	Return: +20 cm after release	**Cartesian line**

Close after descent. At the drop pose:
wait stationary for 1 s, then open and return.

Sources: M1, M3, D2; original image: I2



Recorded green-bowl grasp · 27 Sep 2026

RRTConnect / OMPL

Sample and grow trees from start and goal; connect valid states in six-joint space.

Cartesian interpolation

Sample a tool-space line; solve IK along it. 5 mm step; require 100% path completion.

Validate the full motion before executing it

A candidate is executable only after all six segments pass the planning checks.

CHECK	ACCEPTANCE RULE
Reachability	Valid IK; target inside the workspace
Sampled geometry	Joint bounds; workspace; scene collision checks
Vertical motion	Complete path; bounded joint steps and lateral drift
Endpoint accuracy	Position ≤ 2 mm; orientation ≤ 0.02 rad
Speed and timing	Slow down to limits; reject if duration is excessive
Before each segment	Start-state match; controller readiness; state recheck

PRE-MOTION SELECTION

Try another IK solution
for the same grasp pose.

If full planning fails,
try the next grasp candidate.

Accept one complete plan before sending motion.

EXECUTE Timed joint trajectory → MoveIt ExecuteTrajectory → Robot trajectory controller

Sources: M1–M4 | Values shown are configured acceptance tolerances.